



Review article

Senescent repair memory in chronic disease: Failed resolution as a driver of persistent organ dysfunction

Hongtu Hu^{a,b,*}, Rui Ji^{b,1}, Jijia Hu^{a,**}^a Division of Nephrology, Renmin Hospital of Wuhan University, 238 Jiefang Rd, Wuhan, Hubei, 430060, China^b Department of Nephrology, Chongqing Key Laboratory of Prevention and Treatment of Kidney Disease, Chongqing Clinical Research Center of Kidney and Urology Diseases, Xinqiao Hospital, Army Medical University (Third Military Medical University), Chongqing, China

ARTICLE INFO

Keywords:

Cellular senescence
Senescent repair memory
Chronic disease
Tissue resolution

ABSTRACT

Chronic diseases can continue to progress after the initiating insult has subsided, indicating that injured tissues may retain a persistent imprint of failed repair. Here, senescent repair memory refers to a tissue state in which transient repair-associated senescence fails to resolve and is maintained by persistent senescent cells, altered secretory and metabolic programmes, defective immune clearance, epigenetic stabilization and spatial niche interactions. Once established beyond the adaptive repair window, this state can perpetuate inflammation, fibrosis and functional decline even after the original injury has diminished. This Review examines how failed resolution may contribute to chronic kidney disease, cardiovascular disease, pulmonary fibrosis, metabolic liver disease, neurodegeneration and osteoarthritis. Evidence is strongest in kidney disease, pulmonary fibrosis and metabolic liver disease. Cardiovascular disease and osteoarthritis are supported mainly by experimental or associative studies, whereas the application of this model to neurodegeneration remains largely indirect. Senescent repair memory is therefore presented not as a new molecular pathway or a universal explanation for chronic disease, but as a testable framework linking the persistence, maintenance, spatial organization and functional consequences of unresolved senescence. By distinguishing adaptive repair from disease-sustaining senescence, this perspective may inform the development of stage- and tissue-specific biomarkers and interventions.

1. Introduction

Chronic disease is often framed as the cumulative consequence of persistent injurious stimuli, yet many organs continue to deteriorate after the initiating trigger has been reduced, removed or clinically controlled (Martin et al., 2024; Yeh et al., 2024). This dissociation between insult and progression is evident after acute kidney injury, myocardial infarction, epithelial lung injury, metabolic liver injury and cartilage trauma, where repair programmes can remain active long after structural damage should have resolved (Muto et al., 2024; Tsukui et al., 2024). The expanded hallmarks of ageing now place cellular senescence, chronic inflammation, mitochondrial dysfunction, disabled macroautophagy and dysbiosis within an interconnected network rather than a set of independent degenerative processes (López-Otín et al., 2023; Guo et al., 2022). This network view matters because senescent cells are

active signalling nodes that remodel neighbouring cells, extracellular matrix and immune tone (Huang et al., 2022; Wang et al., 2024a).

This Review examines chronic disease progression as a failure of tissue resolution, in which a transient injury-induced senescence programme persists beyond its physiological role. We define senescent repair memory as a self-maintaining tissue state in which unresolved senescent cells, their secretory and metabolic phenotypes, immune interactions and spatial microenvironment preserve the functional consequences of prior injury. The term memory is used operationally: it refers to persistence of biological information at the level of a tissue system rather than to a discrete storage molecule or a newly identified molecular pathway. Its application requires six linked features: a demonstrable cell-resolved senescence programme, persistence beyond a prespecified adaptive repair window, an identifiable maintenance mechanism, cell-type or spatial context, functional coupling to organ

* Corresponding author at: Division of Nephrology, Renmin Hospital of Wuhan University, 238 Jiefang Rd, Wuhan, Hubei 430060, China.

** Corresponding author.

E-mail addresses: 2014302180179@whu.edu.cn (H. Hu), hujijia@whu.edu.cn (J. Hu).¹ contributed equally to this work

dysfunction and at least partial reversibility after a mechanistically matched perturbation. By contrast, chronic remodelling driven entirely by ongoing injury, matrix accumulation or inflammatory organization without a senescence-dependent component does not meet this definition. The framework therefore organizes established mechanisms into a testable tissue-level model rather than renaming all chronic remodelling as memory.

To translate this definition into a temporal sequence, the repair-memory model can be viewed as four connected stages. Injury first induces a transient senescence programme that limits propagation of damaged cells and coordinates repair through a context-dependent secretory response. Successful resolution then removes or deactivates senescent cells and restores metabolic, immune and stromal homeostasis. When senescent-cell clearance, metabolic recovery, immune surveillance, epigenetic resetting or niche remodelling fails, the programme persists beyond the expected repair period. Reciprocal signalling among senescent cells, immune cells, stromal cells and extracellular matrix can subsequently stabilize a spatially organized niche that sustains organ dysfunction. Senescent repair memory is supported only when these features persist together and respond to selective disruption of the senescent or resolution-maintaining component. The model is weakened or rejected when control of the principal non-senescent driver fully normalizes the tissue phenotype or when selective senescence-directed perturbation changes markers without improving architecture or function.

Viewed in this way, the model reconciles the apparently conflicting observations that senescence can support development, wound repair and tumour suppression while also promoting chronic inflammation, fibrosis and organ decline (Huang et al., 2022; Nakano and Johmura, 2025). The initiating response is not intrinsically pathological; its consequences depend on duration, cellular context and resolution (Martin

et al., 2024; Majewska and Krizhanovsky, 2025). Persistent senescence becomes pathogenic when incomplete clearance or recurrent stress couples the programme to metabolic, immune, epigenetic and spatial feedback that sustains tissue dysfunction. This transition from adaptive repair senescence to pathological repair memory is summarized in Fig. 1.

2. Conceptual framework: from transient senescence to repair memory

Building on this sequence, the pathological significance of cellular senescence depends not only on its presence but also on whether the associated repair programme resolves. The senescent repair memory hypothesis makes three linked claims. First, acute senescence can be adaptive when temporally restricted. Second, chronic pathology can arise when the programme persists after its repair function has ended. Third, persistence is maintained by metabolic, immune, epigenetic and spatial feedback rather than by cell-cycle arrest alone (Huang et al., 2022; Majewska and Krizhanovsky, 2025). The relevant unit of analysis is therefore the unresolved repair trajectory rather than the senescent cell in isolation. A high senescence score does not establish causality unless the implicated compartment is temporally persistent, cell-type resolved, mechanistically maintained, spatially organized and functionally linked to impaired resolution (Suryadevara et al., 2024; Zhao et al., 2024).

This temporal emphasis also changes how senescence should be identified. Cellular senescence is a stress response characterized by durable proliferative arrest, resistance to apoptosis, chromatin remodeling, metabolic adaptation and acquisition of a senescence-associated secretory phenotype (SASP), but none of these features alone defines senescence across tissues (Wang et al., 2024a; Suryadevara et al., 2024).

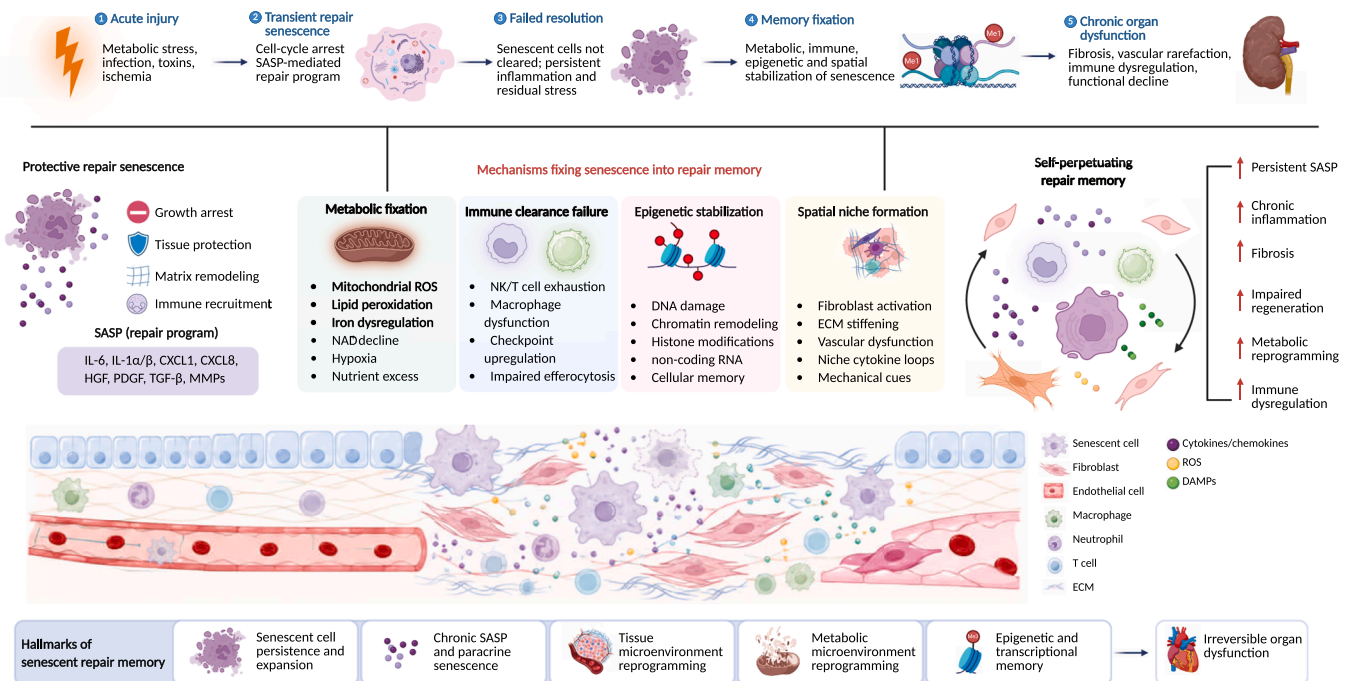


Fig. 1. The senescent repair memory model. Acute tissue injury caused by metabolic stress, infection, toxins or ischaemia induces a transient repair-associated senescence programme characterized by cell-cycle arrest, senescence-associated secretory phenotype (SASP)-mediated signalling, immune recruitment, tissue protection and matrix remodelling. Under physiological conditions, this programme is temporally restricted and contributes to damage containment and repair. When resolution fails, senescent cells are not efficiently cleared and residual inflammatory or metabolic stress persists. Metabolic fixation, impaired immune clearance, epigenetic stabilization and spatial niche formation then convert adaptive senescence into a self-maintaining repair-memory state. This pathological niche sustains chronic SASP signalling, paracrine senescence, metabolic reprogramming, immune dysregulation, fibrosis, impaired regeneration and progressive organ dysfunction. The figure illustrates that chronic disease progression depends not only on senescent-cell accumulation but also on stabilization of unresolved senescence within a disease-perpetuating tissue microenvironment. SASP, senescence-associated secretory phenotype; ECM, extracellular matrix; ROS, reactive oxygen species; DAMPs, damage-associated molecular patterns.

The SenNet recommendations therefore emphasize combinatorial detection across cell-cycle arrest, DNA damage, lysosomal activity, secretory factors, morphology and tissue context rather than reliance on single markers such as p16, p21 or SA- β -galactosidase (Suryadevara et al., 2024; Sanborn et al., 2025). In acute injury, senescence can act as a protective repair programme by limiting propagation of damaged cells, coordinating inflammatory recruitment, promoting matrix remodelling and allowing progenitor-driven restoration of tissue architecture (Martin et al., 2024; Huang et al., 2022). This beneficial phase is most apparent when senescent cells are transient, when their secretory programme recruits immune clearance and when the tissue microenvironment supports regeneration rather than fibrosis (Tsukui et al., 2024; Majewska and Krizhanovsky, 2025).

Beyond cell-cycle arrest, the secretory programme of senescent cells is better viewed as a context-dependent communication system than as a uniformly toxic inflammatory output (Wang et al., 2024a; Majewska and Krizhanovsky, 2025). Early repair-associated SASP factors can recruit macrophages, natural killer cells and T cells, whereas persistent SASP signals can amplify paracrine senescence, endothelial dysfunction, fibroblast activation and extracellular matrix deposition (Majewska and Krizhanovsky, 2025; Majewska et al., 2024). This duality is particularly relevant for chronic disease because interventions that remove

senescent cells too early could impair wound closure, epithelial repopulation or scar stabilization (Huang et al., 2022; Mylonas and Ferenbach, 2024). Conversely, failure to remove senescent cells after the repair phase can lock tissues into a chronic wound-like state that continues to signal damage despite partial restoration of structure (Martin et al., 2024; Muto et al., 2024).

Because the same programme can be beneficial early and harmful when sustained, the temporal boundary between adaptive and maladaptive senescence remains poorly defined in vivo, partly because many experimental studies measure senescence at a single terminal time point (Suryadevara et al., 2024; Melo-Narváez et al., 2024). The relevant repair window is also organ- and model-specific and should therefore be prespecified from the expected kinetics of structural and functional recovery in matched controls rather than inferred retrospectively from marker expression. At minimum, studies should distinguish the injury phase, early repair, late repair and chronic disease, and should determine whether the same cell lineage or state persists rather than attributing a changing bulk signal to cellular memory. Longitudinal single-cell analysis, lineage tracing, serial biofluid measurements and spatial mapping can then establish whether a transient programme resolves, is replaced by a different population or remains fixed within a disease-sustaining niche (SenNet Consortium, 2022; Shao et al., 2024a).

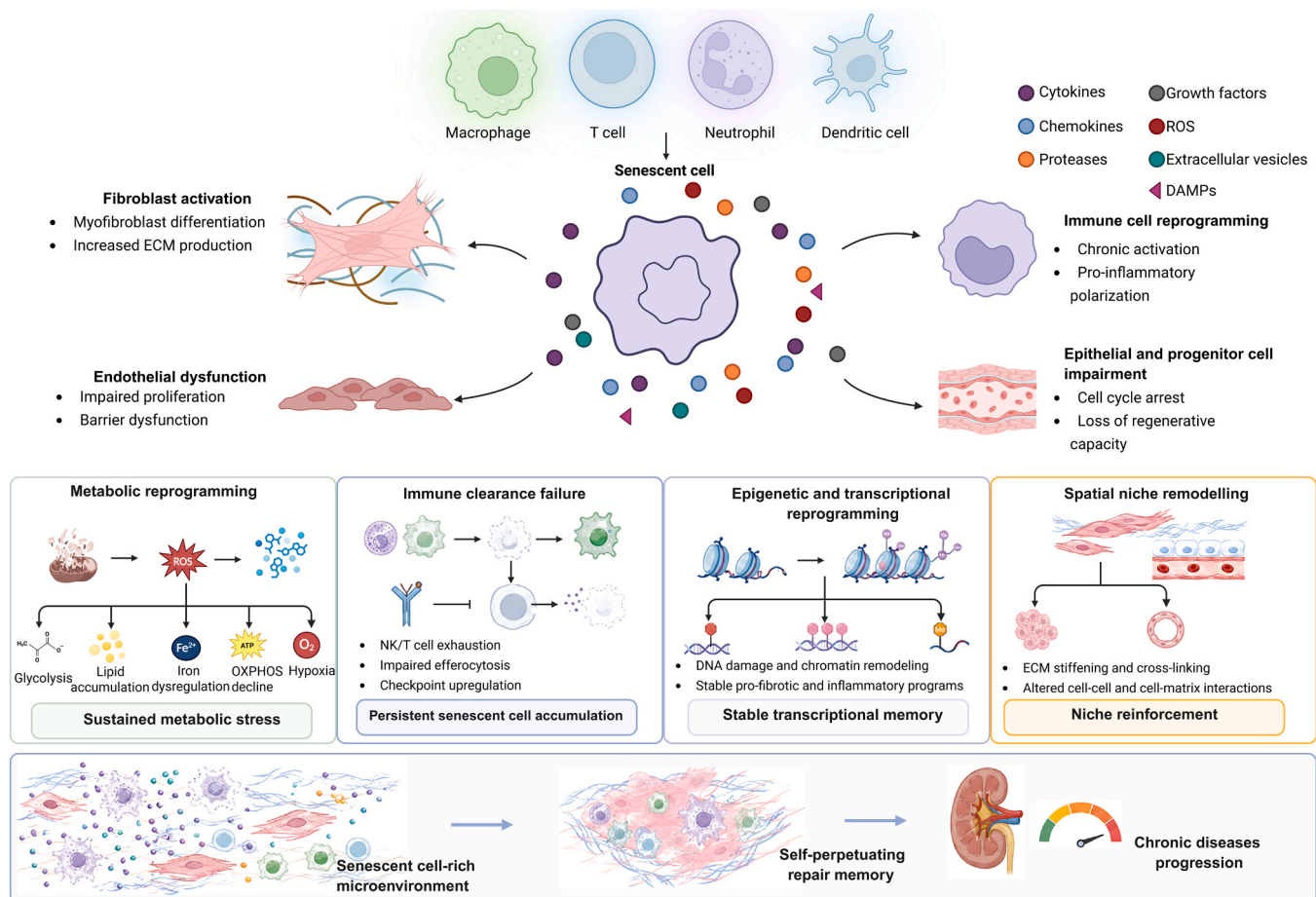


Fig. 2. Mechanisms by which unresolved senescence remodels the tissue microenvironment. Persistent senescent cells act as signalling hubs that reshape neighbouring stromal, immune, endothelial and epithelial compartments through cytokines, chemokines, proteases, growth factors, ROS, extracellular vesicles and DAMPs. These signals promote fibroblast activation and extracellular matrix production, immune-cell reprogramming toward chronic inflammatory states, endothelial dysfunction and epithelial or progenitor-cell impairment. Four interlocking mechanisms stabilize this process. Metabolic reprogramming maintains oxidative stress, lipid accumulation, iron dysregulation, OXPPOS decline and hypoxia. Impaired immune clearance permits senescent-cell accumulation through defective efferocytosis, NK- or T-cell exhaustion and checkpoint upregulation. Epigenetic and transcriptional reprogramming preserves pro-inflammatory and profibrotic gene programmes. Spatial niche remodelling reinforces cell-cell and cell-matrix interactions through ECM stiffening and crosslinking. Together, these mechanisms establish a senescent-cell-rich microenvironment that sustains repair failure and chronic disease progression. OXPPOS, oxidative phosphorylation; NK, natural killer; ECM, extracellular matrix; ROS, reactive oxygen species; DAMPs, damage-associated molecular patterns.

Defining this boundary requires resolution to be treated as an active biological process rather than a passive decline in injury signals. It includes senescent-cell removal, cessation of inflammatory recruitment, normalization of matrix turnover, restoration of parenchymal metabolism and re-establishment of epithelial, endothelial and immune homeostasis (Martin et al., 2024; Muto et al., 2024). Senescent repair memory is proposed as the cellular and tissue correlate of failed resolution, in which coupled senescence-associated programmes alter subsequent tissue behaviour. The state is not assigned to a single molecular mark; it is inferred from persistence of a cell-resolved senescence programme together with metabolic stress, immune retention, epigenetic stabilization and spatially organized stromal interactions (Muto et al., 2024; Vu et al., 2026). These four mechanistic axes reinforce one another and are illustrated in Fig. 2.

These mechanistic requirements can be translated into operational evidence, because senescence burden alone is unlikely to explain disease severity without information on tissue location, senotype, immune visibility and disease stage (Suryadevara et al., 2024; Zhao et al., 2024). A repair-memory interpretation requires evidence that senescence persists beyond the adaptive repair window, is associated with impaired organ function, participates in a stable niche and is at least partly corrected by a senescence-directed or resolution-restoring perturbation. Table 1 summarizes the operational evidence required to support this interpretation, including orthogonal markers in the same cell, longitudinal or stage-resolved sampling, a maintenance mechanism, spatial context and organ-level functional endpoints. These criteria are research standards rather than validated clinical diagnostic cut-offs.

Clinical biopsy studies present a particular challenge because a single specimen cannot by itself distinguish adaptive repair senescence from fixed repair memory. A biopsy can document a senescence-associated tissue state, but the stronger designation of senescent repair memory requires independent temporal or interventional evidence that the state persists beyond the expected repair phase and contributes to organ

dysfunction. Detailed marker panels, quantitative reporting rules and procedures for defining study-specific thresholds are provided in 5 and Table 1. This separation prevents a conceptual definition from being misapplied as a universal diagnostic cut-off.

Such caution is essential because chronic inflammation can occur without a dominant senescent compartment, and senescent cells can be present without driving organ decline (Wang et al., 2024a; Suryadevara et al., 2024). Persistence and spatial organization are not specific to this model and can also characterize myofibroblast expansion, tertiary lymphoid structures or mechanically stabilized fibrosis. Senescent repair memory is supported only when a cell-resolved senescence programme, a maintenance mechanism and a repair-linked functional consequence co-localize, and when selective perturbation of the senescent component improves the phenotype after the principal non-senescent driver has been controlled. A myofibroblast-rich niche should not be classified as repair memory merely because p16 or SASP transcripts are detectable. The senescent population must make an independent, experimentally demonstrable contribution to niche persistence or impaired resolution. Conversely, if correction of the haemodynamic, infectious, metabolic or fibrogenic driver eliminates the phenotype without directly altering the senescent compartment, the repair-memory interpretation is not required. These boundaries keep the concept mechanistically discriminating and falsifiable.

Within these boundaries, the model yields four linked and falsifiable predictions. First, pathological senescence should persist beyond the prespecified repair window and predict subsequent functional decline more closely than the magnitude of the initiating injury (Yeh et al., 2024; Muto et al., 2024). Second, senescence-marker-positive cells should occupy defined immune–stromal, vascular or fibrotic niches and exhibit a maintenance programme that cannot be explained by neighbouring non-senescent drivers alone (SenNet Consortium, 2022; Vu et al., 2026). Third, lineage-restricted or state-selective disruption of the senescent compartment should improve tissue architecture or function,

Table 1
Operational evidence criteria for senescent repair memory.

Evidence domain	Core question	Evidence required	Common misclassification avoided
Temporal persistence	Does senescence remain after the adaptive repair window has passed?	Prespecified time-course sampling covering injury, early repair, late repair and chronic disease. The expected repair window should be derived from matched controls. Persistence must be demonstrated in the same lineage or cell state using an orthogonal marker combination. Report duration and time-point-specific effect sizes. No universal clinical time cut-off is validated.	Calling a transient p21-positive wound edge a chronic disease driver.
Cell identity	Which cell type carries the unresolved programme?	Multiplex or single-cell detection of at least two orthogonal features in the same cell. Suitable combinations include p16 or p21 with reduced Ki-67 or Lamin B1, persistent γ -H2AX or SA- β -gal. Report continuous marker intensity and the proportion of cells meeting the prespecified combination. Define within-study positivity prospectively from negative controls, assay reproducibility and matched control distributions.	Interpreting bulk p16, p21 or SASP signals as a single pathogenic population.
Mechanistic fixation	Which process stabilizes the senescent state?	Quantify maintenance signals and link them to the same cells or niche. Candidate measures include SASP factors such as IL-6, CXCL8, CCL2 or MMPs, mitochondrial or redox stress assessed by ROS, membrane potential or 4-HNE, and immune-retention markers including PD-L1, CD47 or HLA-E. Perturbation of the proposed mechanism should alter the senescent state.	Treating senescence as a generic ageing marker without a maintenance mechanism.
Spatial organization	Is the senescent compartment positioned to influence repair?	Use prespecified spatial transcriptomics, multiplex imaging or neighbourhood analysis. Report enrichment ratios, distance or proximity metrics, or compartment-specific abundance relative to matched non-lesional or control tissue. Diffuse marker positivity alone is insufficient.	Equating dispersed marker positivity with a disease-organizing niche.
Functional coupling	Does the senescent niche track organ dysfunction?	Demonstrate longitudinal association with organ-specific slopes or endpoints, including eGFR slope, fibrosis, ventricular remodelling, lung compliance or cartilage loss. Report effect sizes and sensitivity across plausible study-specific cut-offs. Ideally test a cell-state-selective perturbation after controlling the principal non-senescent driver.	Assuming causality from marker abundance alone.
Reversibility	Can the tissue re-enter a resolution trajectory?	The intervention should reduce the cell-resolved state and improve tissue architecture or organ function. Follow-up after treatment withdrawal should assess recurrence of the niche. Marker reduction alone is not sufficient.	Overstating marker reduction as true disease resolution.

Note: These are research evidence anchors, not validated clinical diagnostic cut-offs. Findings should be reported in three tiers: (1) a senescence-associated state, (2) a persistent senescent niche and (3) senescent repair memory. The strongest designation requires persistence beyond the expected repair phase together with functional coupling or reversibility. A single biopsy cannot distinguish adaptive from fixed repair senescence on marker intensity alone. Any positivity threshold remains assay- and cohort-specific until externally validated.

whereas reduction of an isolated marker without functional recovery should be insufficient (Mylonas and Ferenbach, 2024; McHugh et al., 2025). Fourth, reintroduction or restoration of the relevant maintenance signal should reinstate the pathological phenotype, providing a mechanistic rescue test. Appropriate negative controls include a transient p21-positive wound edge, a bulk SASP signature without cell-type resolution, an ageing-associated inflammatory profile without a persistent repair trajectory and a chronic lesion that resolves fully when the non-senescent driver is removed (Suryadevara et al., 2024; Ogrodnik et al., 2024). Experimental designs should therefore combine temporal tracking, cell-specific perturbation, control of competing drivers and organ-level outcomes rather than rely on cross-sectional association (Muto et al., 2024; Muthamil et al., 2024).

3. Mechanisms that stabilize senescent repair memory

Metabolic stress is a major force that can convert repair-associated senescence into a persistent disease state. Nutrient excess, hypoxia, mitochondrial dysfunction, redox imbalance and lipid peroxidation can continuously reinforce DNA damage responses and inflammatory signalling (López-Otín et al., 2023; Wei et al., 2025). In diabetes, obesity, chronic kidney disease and metabolic liver disease, senescent cells are exposed not to a brief injury pulse but to repeated metabolic insults that can stabilize secretory and anti-apoptotic programmes (Monroe et al., 2025; Sanfeliu-Redondo et al., 2024). Mitochondria are central to this fixation process because mitochondrial dysfunction can promote senescence through impaired bioenergetics, reactive oxygen species generation, altered metabolite pools, mtDNA release and innate immune activation (Guo et al., 2023; Li et al., 2024a). Cytosolic mitochondrial nucleic acids can activate cGAS-STING signalling and thereby connect organelle damage to type I interferon responses, SASP amplification and postmitotic senescence-like states (Hu et al., 2022; Chen et al., 2024a). Recent data further suggest that mitochondrial RNA leakage, in addition to mitochondrial DNA release, can shape the SASP and sustain unresolved injury states (Li et al., 2024a; Victorelli et al., 2025). NAD⁺ decline and defective sirtuin activity may intensify this process by weakening mitochondrial maintenance, DNA repair and inflammatory restraint (Ye et al., 2024; Abdulkhalik et al., 2025).

Within this metabolic context, lipid peroxidation is emerging as a relevant link between metabolic disease and senescence because electrophilic lipid products such as 4-hydroxynonenal can induce genotoxic stress, mitochondrial dysfunction, anti-apoptotic adaptation and senescence in human and murine cells (Monroe et al., 2025). This mechanism may help explain why obesity, MASH, atherosclerosis and diabetic complications share features of oxidative stress, senescence, inflammation and tissue remodelling (Guzmán et al., 2024; Mury et al., 2024). Metabolic phenotypes describe the distinct functional forms of senescence imposed by different metabolic environments (Sanborn et al., 2025; Monroe et al., 2025). Hyperglycaemia may bias senescence toward inflammatory and endothelial-damaging programmes, lipotoxicity toward oxidative lipid-stress responses, hypoxia toward fibrogenic remodelling, and uremic toxins toward premature renal ageing and matrix expansion (Xie et al., 2024; Wu et al., 2024a). Once local senescent niches have formed, normalization of a single systemic variable may be insufficient to reverse them (Katsuumi et al., 2024; Yesilyurt-Dirican et al., 2025). SGLT2 inhibition has reduced senescence-related signals in experimental settings, potentially through metabolic and immune effects, but these observations neither establish SGLT2 inhibitors as senolytics nor prove that their clinical benefits are mediated primarily through senescence resolution (Katsuumi et al., 2024; Yesilyurt-Dirican et al., 2025).

Metabolic fixation is further reinforced by autophagic and lysosomal stress. Damaged organelles, oxidized proteins and undegraded macromolecules can maintain danger signalling even after the initiating stress has diminished (López-Otín et al., 2023; Wang et al., 2024a). Disabled macroautophagy is now recognized as a core ageing hallmark, and its

convergence with mitochondrial dysfunction helps explain why chronically injured tissues often retain a senescence-like transcriptional programme after apparent histological recovery (López-Otín et al., 2023; Suryadevara et al., 2024). Proteostatic stress can also convert a short-lived repair signal into a persistent secretory state because unfolded protein responses, ER stress, lysosomal membrane damage and impaired mitophagy sustain inflammatory and metabolic SASP components (Wang et al., 2024a; Ogrodnik et al., 2024).

These observations shift the therapeutic interpretation of metabolic intervention. Metabolic senomorphics are best considered niche-modifying interventions that reduce the biochemical conditions maintaining persistent senescence rather than generic anti-ageing agents. NAD⁺ restoration, enhancement of mitochondrial quality control, AMPK activation or suppression of lipid peroxidation may modify disease even without directly eliminating senescent cells. SGLT2 inhibitors and GLP-1-based therapies should primarily be interpreted through their established metabolic, haemodynamic and direct parenchymal actions; any senescence-modulating or resolution-promoting effect remains a context-dependent pharmacodynamic hypothesis requiring cell-specific target engagement and organ-level validation (Katsuumi et al., 2024; Yesilyurt-Dirican et al., 2025; Drucker, 2025).

Metabolic fixation is only one component of persistence. Physiological senescence is also coupled to immune surveillance, and senescent cells can be eliminated by macrophages, natural killer cells, CD8⁺ T cells and other immune effector populations when their immunogenic signals remain intact (Majewska and Krizhanovsky, 2025; Wang and Nakanishi, 2025). Failure of this clearance machinery can transform a transient repair population into a persistent source of SASP, paracrine senescence and local immune distortion (Majewska and Krizhanovsky, 2025; Ajoalabady et al., 2024). The ageing immune system is not simply weaker; it is remodelled toward chronic inflammation, clonal skewing, exhaustion, impaired phagocytosis and reduced regenerative support (Delgado-Pulido et al., 2025; Bracken et al., 2025). These changes can reduce the ability of tissues to clear senescent cells while simultaneously increasing the inflammatory signals that induce new senescence (Majewska and Krizhanovsky, 2025; Ajoalabady et al., 2024). Senescent cells can also evade clearance by upregulating inhibitory immune checkpoints, thereby acquiring a form of immune escape that resembles tumour immunology but operates in non-malignant chronic disease (Majewska et al., 2024; Salminen, 2024). p16-dependent stabilization of PD-L1 in senescent cells supports their accumulation in ageing and chronic inflammation, while PD-L1 downregulation can enhance immune-mediated senescent-cell removal (Majewska et al., 2024; Katsuumi et al., 2024).

Senescent cells can further evade immune surveillance through HLA-E-NKG2A, CD47-SIRPα, other anti-phagocytic ligands and the ganglioside GD3, each of which may promote persistence within injured tissues (Salminen, 2024; Iltis et al., 2025). These mechanisms indicate that repair memory can be maintained by immune tolerance as well as by ongoing injury (Majewska and Krizhanovsky, 2025; Iltis et al., 2025). In preclinical models, a chimeric peptide that enhances immune recognition of senescent cells improves outcomes across injury, fibrosis, tumourigenesis and ageing, whereas uPAR-directed senolytic CAR T cells provide sustained metabolic and functional benefits in aged or high-fat-diet-exposed mice (Ming et al., 2025; Amor et al., 2024). These findings support immune-guided clearance as a potentially more selective approach than indiscriminate senolysis, although tissue specificity and preservation of adaptive repair remain essential (Majewska and Krizhanovsky, 2025; McHugh et al., 2025).

At the tissue level, immune retention develops when surveillance cells are chronically reprogrammed by SASP factors, metabolic waste, immune checkpoints and extracellular-matrix barriers (Majewska and Krizhanovsky, 2025; Zhou et al., 2024). Macrophages, natural killer cells and cytotoxic T cells can then shift from clearance toward tolerance, while impaired efferocytosis allows senescent material and inflammatory signals to persist (Majewska et al., 2024; Zhou et al., 2024).

Immune escape by senescent cells and functional reprogramming of immune cells therefore form a reciprocal loop that stabilizes the niche rather than merely reflecting generalized immunosenescence.

This reciprocal immune loop also provides a context for interpreting experimental canagliflozin studies, which illustrate how a metabolic intervention may indirectly enhance senescent-cell clearance by reducing PD-L1-associated immune evasion (Majewska et al., 2024; Katsuumi et al., 2024). This interpretation is mechanistically consistent with evidence that PD-L1-positive senescent cells can resist T-cell surveillance and that PD-1 blockade can promote their removal in vivo (Wang et al., 2022). These findings do not establish immune-mediated senescence clearance as the principal clinical mechanism of SGLT2 inhibitors. They instead identify a testable pathway through which selected metabolic therapies could influence tissue resolution. Effective immune-guided senotherapy is likely to require improved recognition of pathological senescent cells, restored efferocytosis and relief of SASP-driven immune dysfunction (Wang et al., 2024a; Majewska and Krizhanovsky, 2025).

Alongside metabolic and immune fixation, persistent senescence is reinforced by chromatin reorganization, heterochromatin instability, DNA damage signalling, histone modification and altered enhancer accessibility (Muto et al., 2024; Wu et al., 2024b). These changes can preserve injury-induced transcriptional states after the original stress has declined, providing a molecular substrate for repair memory (Muto et al., 2024; Wu et al., 2024b). In the kidney, single-nucleus multiomic profiling of successful and failed repair after acute injury has shown that maladaptive repair involves sustained epigenetic reprogramming in injured epithelial and stromal compartments (Muto et al., 2024). More broadly, chronic organ dysfunction may reflect the persistence of injury-accessible chromatin states even after the initiating damaging agent has diminished (Yeh et al., 2024; Muto et al., 2024).

These chromatin changes are themselves coupled to metabolism because mitochondrial metabolites, acetyl-CoA availability, NAD⁺ status, fumarate, succinate and redox balance can modify chromatin-regulating enzymes and transcriptional programmes (Zhang et al., 2024a; Zheng and Yin, 2025). This metabolic-epigenetic coupling may help explain how chronic nutrient stress translates into durable senescence, fibrosis and impaired regenerative capacity (Muto et al., 2024; Monroe et al., 2025). Chromatin memory also has implications for biomarkers because senescent-cell detection based solely on mRNA expression may miss latent transcriptional potential encoded in chromatin accessibility (Suryadevara et al., 2024; Sanborn et al., 2025). Multiomic senescence maps that combine transcriptomic, epigenomic, proteomic and spatial features are therefore likely to be more informative than single-marker assays for identifying senescent repair memory (Muto et al., 2024; SenNet Consortium, 2022).

When metabolic, immune and epigenetic programmes become spatially organized, senescent cells exert their strongest pathological effects when they are spatially positioned to influence regenerative, immune, vascular or stromal compartments (Suryadevara et al., 2024; SenNet Consortium, 2022). The relevant unit in chronic disease may therefore be the senescent niche rather than the isolated senescent cell (Tsukui et al., 2024; Vu et al., 2026). A senescent niche can be defined as a spatially organized microenvironment in which senescent cells interact with fibroblasts, macrophages, endothelial cells, nerves, progenitors and extracellular matrix to maintain inflammation and failed repair (Tsukui et al., 2024; Vu et al., 2026). Such niches can produce self-reinforcing feedback because matrix stiffness promotes senescence, senescent cells remodel matrix, and altered matrix then constrains tissue regeneration (Zhao et al., 2024; Fu et al., 2024). Spatial transcriptomic analyses in human MASLD have linked fibrosis progression to coordinated zones of metabolic reprogramming and senescence, illustrating how disease-relevant senescence can be embedded in tissue architecture (Vu et al., 2026). In fibrotic lung injury, specialized alveolar fibroblast lineages sequentially adopt inflammatory and profibrotic states that provide or distort epithelial stem-cell niches (Tsukui et al., 2024).

The resulting spatial organization can amplify chronic disease rather than simply mark it (Tsukui et al., 2024; Vu et al., 2026). This interpretation supports local or niche-targeted interventions instead of systemic depletion of all senescent cells (Magkouta et al., 2025; Gu et al., 2024). Spatial context is also essential for biomarker interpretation because the same marker may have different implications in epithelial cells, fibroblasts, endothelial cells, macrophages or progenitor compartments (Suryadevara et al., 2024; Zhao et al., 2024). Studies should therefore report cell identity, distance to fibrotic matrix, proximity to immune cells, vascular association and relationship to regenerative zones in addition to marker expression (Sanborn et al., 2025; SenNet Consortium, 2022). Extracellular matrix structure is likely to be a major physical component of senescent repair memory because matrix stiffness, collagen crosslinking and altered basement membranes can preserve aberrant mechanotransduction after the original injury has subsided (Martin et al., 2024; Diekmann and Loeser, 2024).

Matrix remodelling can then provide a durable physical component of repair memory and may explain why some tissues remain locked in a profibrotic or degenerative state despite reduction of inflammation, because the matrix itself continues to instruct epithelial, endothelial, immune and stromal cells to behave as though injury is ongoing (Fu et al., 2024; Diekmann and Loeser, 2024). Spatial transcriptomics adds resolution to this concept by showing that metabolically, inflammatory, senescence and fibrotic signatures can be topologically linked within diseased human tissue rather than distributed uniformly across an organ (Suryadevara et al., 2024; Vu et al., 2026). Consequently, the most informative studies will integrate multiplex imaging, spatial transcriptomics, single-cell multiomics and functional perturbation to identify which senescent neighbourhoods are drivers, bystanders or adaptive repair zones (Suryadevara et al., 2024; Ogrodnik et al., 2024).

Together, these measurements can prevent a high senescence score in bulk tissue from being misinterpreted as evidence of a pathogenic senescent niche when the signal instead reflects cell composition, sampling location or repair stage (Suryadevara et al., 2024; Muthamil et al., 2024). They also provide a common basis for comparing how strongly the repair-memory model is supported across different organs.

4. Disease contexts that test the repair-memory model

Using these common criteria, evidence for senescent repair memory varies substantially across organs and should be graded rather than presented as uniform. In this Review, the strength of support is judged across five domains: temporal persistence beyond an expected repair period, orthogonal cell-type-resolved evidence of senescence, spatial linkage to a disease-relevant niche, mechanistic evidence for maintenance and functional improvement after selective perturbation. Human longitudinal or interventional data provide the strongest support, whereas cross-sectional marker associations without cell identity or functional validation provide only indirect evidence. Kidney disease, pulmonary fibrosis and MASLD currently show the most coherent convergence across these domains, although causal human evidence remains incomplete. Cardiovascular disease and osteoarthritis provide substantial experimental support but less consistent human evidence for a disease-organizing senescent niche. Neurodegeneration remains an emerging and more indirect test because postmitotic biology, marker heterogeneity and limited longitudinal human sampling complicate interpretation. To maintain comparability, each disease context is considered using the same questions: which injury initiates the state, which cellular compartment carries it, which niche or maintenance mechanism stabilizes it, how strong the human evidence is and which experiment could reject the repair-memory interpretation. Across all settings, the critical question is not simply whether senescent cells are present, but whether injury-induced senescence persists, is mechanistically maintained and remains spatially and functionally coupled to chronic dysfunction (Zhao et al., 2024; Ogrodnik et al., 2024).

4.1. Kidney and cardiovascular repair memory

Among the available organ models, chronic kidney disease (CKD) provides the strongest organ-specific test of the framework because failed epithelial repair can be followed from AKI to chronic fibrosis and functional decline (Yeh et al., 2024; Muto et al., 2024). Tubular epithelial cells, podocytes, endothelial cells, macrophages and fibroblasts can acquire senescence-related features, but their contributions depend on injury timing, cell type and microenvironment (Zhang et al., 2024a; Hejazi et al., 2024; Chen et al., 2024b; Rao et al., 2024). Conditional proximal-tubule models and studies of uremic or metabolic stress support a role for persistent tubular and stromal states in local inflammation and fibrosis, while AhR activation and impaired Klotho signalling provide plausible metabolic maintenance mechanisms (Mylonas and Ferenbach, 2024; Shao et al., 2024a; Xie et al., 2024; Zhang et al., 2024b). Human biomarker studies, including CKAP4 and PTX3, remain supportive but not specific for a renal senescent niche (McCallion et al., 2024). Mosaic loss of the Y chromosome provides an additional example of genomic mosaicism associated with ageing and epithelial injury in CKD (Wilson et al., 2024). These findings support hypothesis-driven approaches to kidney ageing, but clinical biopsy criteria and causal intervention studies are still needed (Russo et al., 2024).

Within the kidney, macrophage dysfunction adds an immune-metabolic layer to renal ageing and fibrosis (Wu et al., 2024a). Single-cell and experimental studies suggest that macrophage–tubule communication, iron dyshomeostasis and impaired immune resolution can reinforce epithelial and interstitial pathology (Wu et al., 2024a). A repair-memory interpretation therefore requires renal cell-type-resolved senescence markers, spatial immune–stromal mapping and repeated functional measurements rather than bulk inflammatory or senescence scores. Kidney evidence is more developed than that in several other organs, but it does not yet support a validated clinical diagnostic framework (Muto et al., 2024; Zhang et al., 2024a; McCallion et al., 2024).

These interactions are particularly amenable to longitudinal study in the AKI-to-CKD transition, which provides a defined temporal model because tubular cells that fail to redifferentiate may remain in a maladaptive secretory and profibrotic state after the acute insult has ended (Yeh et al., 2024; Muto et al., 2024). Short-range communication among tubular, glomerular, vascular and interstitial compartments can then sustain injury (Suryadevara et al., 2024; Wu et al., 2024a). In diabetic kidney disease, hyperglycaemia, lipotoxicity, hypoxia, iron dyshomeostasis and uremic solutes may impose overlapping metabolic senotypes on proximal tubules, podocytes and endothelial cells (Xie et al., 2024; Zhang et al., 2024b). A rigorous test therefore requires concordant changes in longitudinal function, segment-resolved markers, spatial niche structure and a mechanistically matched perturbation rather than elevation of a single renal marker.

Even within this model, renal repair memory is likely to vary by disease stage and compartment. Early tubular, podocyte or endothelial stress may be more reversible than advanced interstitial fibrosis, and the same pathway may be adaptive during acute stress but maladaptive when sustained. Studies should therefore pair albuminuria or eGFR slope with cell-resolved senescence markers, spatial immune–stromal mapping and stage-appropriate interventions (Suryadevara et al., 2024; Ogrodnik et al., 2024).

Cardiovascular disease offers a complementary test in a mechanically and metabolically stressed system in which vascular, myocardial and stromal compartments may retain injury-associated states after haemodynamic or ischaemic triggers have changed (Martin et al., 2024; Shen et al., 2024). Senescence-related states in endothelial cells, smooth-muscle cells, fibroblasts, immune cells and perivascular compartments have been linked to vascular dysfunction and cardiac remodelling (Mury et al., 2024; Abdellatif et al., 2023, 2025; Ministrini et al., 2025; Dong et al., 2024; Zhao et al., 2025a; Silva et al., 2025;

Wagner et al., 2023; Vanicek et al., 2025; Xu et al., 2025; Sutanto et al., 2025; Hall and Lesniewski, 2024). Experimental senescent-cell clearance in HFpEF models supports potential causality, but most human studies remain cross-sectional and are confounded by age, haemodynamic load, obesity, diabetes, renal dysfunction and treatment exposure. Consequently, current data do not establish a universal cardiac repair-memory signature or identify which cardiovascular compartment is the dominant causal reservoir in individual patients.

Within the heart, senescent cells may contribute to heart failure by promoting inflammation, endothelial damage, fibrosis and impaired myocardial-stromal communication (Zhao et al., 2025a; Silva et al., 2025). Pharmacological clearance of senescent cells in models of heart failure with preserved ejection fraction (HFpEF) reduces inflammation, endothelial damage and cardiac fibrosis, supporting a causal role for persistent senescence in maladaptive cardiac remodelling (Silva et al., 2025). The cardiac neurovascular interface may represent another spatial component of repair memory because ageing alters nerve density and vascular-derived neuroregulatory programmes in the heart (Wagner et al., 2023). Pericyte senescence in HFpEF further highlights that stromal-vascular support cells can encode chronic dysfunction even when cardiomyocytes are not the primary senescent population (Vanicek et al., 2025).

These findings suggest that cardiovascular repair memory is likely to arise from interactions among endothelial cells, smooth-muscle cells, fibroblasts, immune cells and perivascular niches rather than from a single senescent-cell population (Xu et al., 2025; Sutanto et al., 2025). Stronger human evidence will require cell-resolved myocardial or vascular profiling linked to disease stage, spatial localization of the implicated population, and longitudinal association with ventricular remodelling, vascular stiffness or clinical function. Mechanistic studies should distinguish senescence-related effects from persistent pressure or volume load, ischaemia, sex, metabolic background and immune state, and should test whether selective targeting of the proposed senescent compartment adds benefit after these drivers are controlled. Therapeutic studies must likewise demonstrate tissue-specific target engagement and functional benefit rather than rely on a fall in circulating SASP proteins or systemic senescence scores (Mury et al., 2024; Hall and Lesniewski, 2024).

A recent study further illustrates how multicellular cardiac ageing can be maintained without a single dominant senescent population. In aged mice, macrophage ARG2 promoted IL-1 β -dependent injury across cardiomyocytes, fibroblasts and endothelial cells, whereas fibroblast ARG2 acted through mitochondrial reactive oxygen species (Potenza et al., 2025). This intercellular circuit is compatible with a repair-memory niche, but it does not by itself establish persistence after a discrete injury or selective reversibility. A decisive cardiac test would follow myocardial or vascular cell states after haemodynamic stabilization or infarct healing, map the implicated macrophage-stromal compartment and determine whether stage-specific disruption improves remodelling after the initiating load has been controlled.

4.2. Lung and metabolic liver disease

A second well-developed context is chronic lung disease, where repeated epithelial injury can convert a regenerative programme into a spatially fixed fibrotic state (Martin et al., 2024; Tsukui et al., 2024). The lung is highly vulnerable to senescent repair memory because epithelial regeneration requires coordinated interaction among alveolar type 2 (AT2) cells, fibroblast niches, endothelial cells and immune populations after injury (Tsukui et al., 2024; Wang et al., 2024a). When this coordination fails, epithelial progenitor exhaustion, fibroblast activation and persistent inflammation can convert repair into pulmonary fibrosis or chronic obstructive pulmonary disease (COPD) remodelling (Parimon et al., 2023; Liu et al., 2024). AT2 cell senescence impairs epithelial self-renewal and contributes to fibrotic lung injury (Su et al., 2024; Chin et al., 2023). Loss of IGFBP2 and Lonp1 deficiency both implicate

ageing-associated AT2 dysfunction in susceptibility to fibrosis, reinforcing the view that epithelial repair memory can originate in progenitor-like compartments (Chin et al., 2023; Zhu et al., 2024).

Within this epithelial-niche framework, mitochondrial injury, circadian disruption, oxidative lysosomal damage and hypoxia-related stress can promote alveolar epithelial senescence and fibrotic remodeling (Peng et al., 2024; Chen et al., 2024c). ROS-mediated lysosomal membrane permeabilization and autophagy inhibition regulate bleomycin-induced senescence, linking organelle quality control to persistent pulmonary injury responses (Qi et al., 2024). COPD adds a paracrine and extracellular-vesicle dimension to senescent repair memory because airway epithelial cells from COPD can transfer miR-34a-containing vesicles that induce senescence in healthy recipient cells (Devulder et al., 2025). IFN-mediated epithelial senescence in COPD further suggests that chronic antiviral-like inflammatory programmes can stabilize pathological airway states (Guo-Parke et al., 2025).

Chronic lung disease can therefore be sustained by epithelial senescence, distorted fibroblast niches, immune retention and defective autophagy rather than by ongoing epithelial injury alone (Tsukui et al., 2024; Parimon et al., 2023). Lung fibrosis therefore provides relatively strong experimental and translational support for the model, including epithelial-niche and spatial mechanisms, but human interventional evidence remains limited. Strategies that restore alveolar progenitor niches, reduce pathological SASP and preserve adaptive epithelial repair may be preferable to broad senolysis (Li et al., 2024b; Wang et al., 2024b).

The comparison between idiopathic pulmonary fibrosis and COPD further shows that these diseases test different forms of pulmonary repair memory and should not be treated as a single senescent phenotype. In pulmonary fibrosis, the candidate reservoir is centred on dysfunctional AT2 cells and the fibroblast niche, whereas COPD may involve airway epithelial propagation through extracellular vesicles and interferon-linked paracrine programmes (Parimon et al., 2023; Liu et al., 2024; Su et al., 2024; Chin et al., 2023; Zhu et al., 2024; Peng et al., 2024; Chen et al., 2024c; Qi et al., 2024; Devulder et al., 2025; Guo-Parke et al., 2025). The strongest pulmonary experiment would combine lineage- or state-resolved sampling across injury and recovery, spatial measurement of epithelial-fibroblast and epithelial-immune interactions, and intervention after the acute inflammatory phase. Improvement in compliance, gas exchange or fibrosis after selective disruption would provide stronger evidence than reduced epithelial p16 or SASP expression alone.

The liver provides a different stress test because, in metabolic dysfunction-associated steatotic liver disease (MASLD), the initiating stress is systemic and nutrient-driven, while pathological memory becomes organized within local hepatocyte, immune and stellate-cell niches (Vu et al., 2026; Monroe et al., 2025). In metabolic dysfunction-associated steatohepatitis (MASH), nutrient stress, lipid toxicity and chronic inflammation can fix senescence into fibrotic repair memory (Sanfeliu-Redondo et al., 2024; Li and Wang, 2024). Human and murine MASH studies show that hepatic stellate-cell senescence is heterogeneous, stage-dependent and intertwined with fibrogenesis and immune interactions (Yashaswini et al., 2024). Single-nucleus analyses have defined senescent hepatic stellate-cell phenotypes in human and murine MASH and traced their emergence from activated stellate-cell populations (Yashaswini et al., 2024). The evolving specificity of uPAR expression across stellate and myeloid populations illustrates why senolytic targets can change across disease stage and tissue context (Amor et al., 2024; Yashaswini et al., 2024).

Within this metabolic setting, spatial transcriptomic evidence that progressive MASLD fibrosis is associated with linked metabolic reprogramming and senescence provides one of the strongest human tissue-level supports for the repair-memory model (Vu et al., 2026). In this setting, senescence is not merely a cell-intrinsic arrest state but part of a fibrotic metabolic architecture that organizes inflammatory and stromal

signalling (Vu et al., 2026; Guzmán et al., 2024). Hepatocellular senescence can also induce multi-organ dysfunction, although this inter-organ extension remains mechanistically emerging (Kiourtis et al., 2024). Most liver-directed senolytic and niche-remodelling interventions remain preclinical, so strong spatial association should not be equated with clinical therapeutic readiness (Gu et al., 2024; Yang et al., 2025).

These stage-dependent effects are also reflected in therapeutic strategies, which illustrate the need to distinguish adaptive from maladaptive senescence (Gu et al., 2024; Yang et al., 2025). Targeted manganese delivery to activated hepatic stellate cells combines senescence induction with immune clearance to limit fibrosis, whereas BCL-xL/BCL-2 PROTAC approaches seek to remove senescent liver cells in MASH-driven hepatocarcinogenesis (Gu et al., 2024; Yang et al., 2025).

Accordingly, MASLD demonstrates why senescence cannot be assigned a fixed fibrogenic meaning. Stellate-cell senescence may restrict proliferation in one context yet sustain fibrosis when immune clearance fails or when secretory and matrix-remodelling programmes persist (Yashaswini et al., 2024). Hepatocyte-derived TGF- β can extend senescence-associated dysfunction beyond the liver, but systemic propagation should be distinguished from proof that a local hepatic niche retains the initiating injury (Kiourtis et al., 2024). The most informative studies would align metabolic improvement, spatial hepatocyte-stellate-myeloid states and fibrosis regression over time, then determine whether the same niche re-emerges after treatment withdrawal.

4.3. Neurodegeneration and mechanically stabilized osteoarthritis

Neurodegeneration provides a more stringent test because long-lived postmitotic cells and glial populations can acquire persistent senescence-like inflammatory and metabolic states without fulfilling all classical criteria for proliferative arrest (Suryadevara et al., 2024). An integrative longitudinal single-cell atlas in Alzheimer disease model mice identified an exhausted-like microglial population that increased with ageing and APOE4 burden, with an analogous population detectable in human Alzheimer disease tissue; human spatial transcriptomics independently mapped plaque-associated glial states across vulnerable cortical regions (Millet et al., 2024; Avey et al., 2025). These observations strengthen the cell-state and spatial components of the model, but they also expose an important alternative explanation: exhausted or reactive glia may persist without being senescent. Neurons, astrocytes, microglia and oligodendroglial lineage cells can develop senescence-like features after ageing, proteotoxic stress, mitochondrial injury or inflammatory exposure (Wang et al., 2024c; de Luzy et al., 2024; Rim et al., 2024; Carling et al., 2024). A repair-memory interpretation must therefore distinguish cellular senescence from reactive gliosis, immune exhaustion, terminal differentiation and generic proteostatic stress. Evidence should combine persistent DNA-damage signalling, lysosomal or metabolic alteration, a stable secretory programme, loss of homeostatic function and spatial association with vulnerable neural circuits rather than rely on p16, p21 or inflammatory transcripts alone. Microglial and astrocytic findings remain biologically plausible but incompletely causal because longitudinal human sampling and selective clearance studies are limited (Suryadevara et al., 2024; Wang et al., 2024c).

Within this setting, astrocyte senescence can compromise neuronal support, metabolic coupling and inflammatory restraint (Alshaebi et al., 2025; Caceres-Palomo et al., 2025). Human iPSC-derived APOE4/4 astrocytes exhibit senescent and pro-inflammatory features that impair neuronal support, providing a disease-relevant model of glial repair failure (Caceres-Palomo et al., 2025). Neurodegenerative senescence is also spatial because glial reactivity, synaptic dysfunction and tau- or amyloid-associated microenvironments evolve across vulnerable brain regions (Rohden et al., 2025; Gruel et al., 2024). S100A8-enriched microglia in tauopathy and accelerated ageing further suggest that

disease-associated glial states can converge on inflammatory senescence-like programmes (Gruel et al., 2024).

Spatial evidence adds another layer, as human plaque-glia maps show that local amyloid burden, microglial activation and astrocytic responses vary within and between brains, supporting a niche-based rather than organ-wide interpretation (Avey et al., 2025). However, cross-sectional postmortem spatial association cannot establish that the niche persists beyond a repair window or drives subsequent decline. Stronger evidence would require longitudinal fluid or imaging proxies linked to cell-resolved tissue validation, selective perturbation of the candidate glial state and demonstration that synaptic or behavioural recovery exceeds the effect of lowering amyloid or tau alone.

These uncertainties make therapeutic interpretation particularly cautious. Senolytic approaches are being explored for Alzheimer disease and Parkinson disease, but the brain requires particular caution because glial cells participate in debris clearance, synaptic support, metabolic coupling and tissue repair (Richardson and Richardson, 2024; Garbarino et al., 2025). A reduction in a glial senescence-like signature could therefore be harmful if it also removes cells needed for plaque containment or neuronal support. The repair-memory framework favours interventions that restore microglial and astrocytic homeostatic functions, reduce a demonstrably pathogenic secretory programme and preserve beneficial surveillance rather than indiscriminately deplete all senescence-like glial cells (Rim et al., 2024; Alshaebi et al., 2025). Brain-directed studies should incorporate regional target engagement, neuroinflammatory and synaptic endpoints, cognitive or motor outcomes and delayed safety follow-up, because short-term marker reduction alone cannot establish neurological benefit.

Osteoarthritis presents a different challenge because whole-joint heterogeneity can complicate interpretation. A human single-cell atlas identified homeostatic, proliferative fibrochondrocyte and hypertrophic chondrocyte populations, showing that an apparent senescence signature may partly reflect differentiation state or sampling depth (Huang et al., 2024). Conversely, inhibition of 15-hydroxy prostaglandin dehydrogenase promoted cartilage regeneration in aged and post-traumatic mouse joints and induced regenerative transcriptional changes in existing chondrocytes, with supportive findings in human osteoarthritic cartilage explants (Singla et al., 2026). This work provides a functional example of reversibility of an age-associated cartilage state, but it should not be interpreted as proof that senescent-cell clearance caused repair. A repair-memory test should localize the responsive chondrocyte state, determine whether matrix mechanics or synovial inflammation maintains it and relate local target engagement to structural and pain-related outcomes.

Against this heterogeneous background, osteoarthritis provides a strong experimental example of mechanically stabilized repair pathology because cartilage injury, matrix stiffening and chondrocyte senescence can reinforce one another after the initiating insult has changed (Martin et al., 2024; Fu et al., 2024). Chondrocyte senescence is associated with matrix degradation, inflammatory SASP and impaired cartilage homeostasis, although functional senescence and terminal differentiation remain difficult to separate (Suryadevara et al., 2024; Diekmann and Loeser, 2024). Matrix stiffening and mitochondrial stress provide plausible maintenance mechanisms (Fu et al., 2024; Shao et al., 2024b). The evidence is therefore substantial in experimental models but less mature for a clinically validated, cell-resolved repair-memory definition; local senotherapy remains a hypothesis requiring tissue-stage and safety-specific testing.

At the molecular level, multiple regulators of chondrocyte senescence, including Sirt4, SIRT6 and miR-653-5p, have been implicated in recent osteoarthritis models (Lin et al., 2024a, 2024b; Wang et al., 2024d). HNRNP-mediated RNA regulation has also been linked to chondrocyte senescence and osteoarthritis progression (Jiang et al., 2024). These studies suggest that cartilage senescent repair memory is not a single pathway but a convergent state shaped by mitochondrial metabolism, RNA regulation, epigenetic control and inflammatory

matrix signalling (Fu et al., 2024; Shao et al., 2024b). Senolytic therapy with dasatinib and quercetin can reduce chondrocyte senescence in preclinical or translational osteoarthritis studies, but clinical and tissue-stage specificity remains unresolved (Zhao et al., 2025b; Maurer et al., 2025). Small extracellular vesicle and prodrug strategies that rejuvenate aged chondrocytes or target beta-galactosidase-positive cells may offer more localized approaches to joint senotherapy (Feng et al., 2024; Li et al., 2025).

The osteoarthritis evidence highlights two broader implications: tissue mechanics can stabilize senescence-associated pathology, and senotherapy must be adapted to the structural constraints of the diseased niche (Fu et al., 2024; Zhang et al., 2025). Similar principles may apply to kidney fibrosis, vascular stiffness and pulmonary scarring, in which extracellular matrix changes also preserve maladaptive repair states (Muto et al., 2024; Tsukui et al., 2024).

5. Biomarkers, experimental models and operational evidence criteria

The disease comparisons above highlight why robust identification of senescent repair memory requires biomarkers that capture cell identity, senescence state, metabolic fixation, immune retention and spatial organization (Suryadevara et al., 2024; McCallion et al., 2024). Single markers such as p16, p21, SA- β -galactosidase or γ -H2AX are useful entry points but cannot distinguish adaptive repair senescence from pathological memory in complex tissues (Wang et al., 2024a; Suryadevara et al., 2024). A practical research panel should demonstrate at least two orthogonal senescence features in the same cell. Suitable combinations include p16 or p21 with reduced Ki-67 or Lamin B1, persistent DNA-damage signalling or SA- β -galactosidase, supplemented by context-dependent SASP factors such as IL-6, CXCL8, CCL2 or MMPs. Metabolic and immune-retention readouts may include ROS, mitochondrial membrane potential, 4-HNE, PD-L1, CD47 or HLA-E, interpreted in relation to fibrotic or inflammatory niches (Suryadevara et al., 2024; Monroe et al., 2025; Salminen, 2024; Iltis et al., 2025). Quantification should be prespecified at three levels: continuous marker intensity, the proportion of cells meeting the same-cell marker combination and spatial enrichment relative to matched non-lesional or control tissue. Within-study positivity thresholds should be derived prospectively from negative controls, assay reproducibility and matched control distributions. When longitudinal outcomes are available, performance should be examined across plausible cut-offs rather than selecting a single threshold post hoc. Effect sizes and sensitivity analyses should accompany any dichotomized result. Because assay platforms, tissue composition and repair kinetics differ among organs, such thresholds remain study-specific and require external validation before clinical use.

To test these criteria experimentally, models should capture the transition from acute injury to chronic dysfunction rather than only terminal disease (Yeh et al., 2024; Muto et al., 2024). Time-course models, aged animals, metabolic-stress models, senescence reporter systems, lineage tracing, organoids and spatial omics each illuminate different layers of repair memory (Melo-Narváez et al., 2024; Shao et al., 2024a). A rigorous design should include an early adaptive phase, an expected resolution phase and a late persistent phase; identify the same cell lineage across time; and combine molecular markers with tissue architecture and organ function. Conditional induction or ablation should be performed at more than one stage to determine whether the same senescent population is beneficial early but pathogenic later. Rescue experiments should then restore the proposed maintenance signal or reconstitute the implicated cell-cell interaction. Human validation is particularly important because rodent senescence programmes, injury kinetics and immune surveillance may not fully reproduce human chronic disease (Suryadevara et al., 2024; Karpova et al., 2026). Integrating human tissue atlases, plasma proteomics, organ-specific ageing clocks and longitudinal clinical phenotyping could connect senescent

repair memory to disease trajectory and therapeutic windows (Shen et al., 2024; Wang et al., 2026). The minimum-information approach for in vivo senescence experiments is especially relevant because it discourages reliance on a single marker and requires combinations of cell-cycle arrest, molecular damage, secretory phenotype, morphology and functional evidence (Suryadevara et al., 2024; Ogrodnik et al., 2024).

Even with rigorous experimental models, circulating biomarkers in translational studies should be interpreted as entry points rather than substitutes for tissue evidence, because plasma SASP proteins may reflect multiple organs, immune activation or comorbidity rather than a specific senescent niche (Muthamil et al., 2024; Carver et al., 2025). IL-23R has recently emerged as a senescence-linked circulating and tissue biomarker of ageing, but its greatest value for chronic disease may lie in combination with organ-specific injury markers and spatial tissue validation (Suryadevara et al., 2024; Carver et al., 2025). Organ-specific proteomic ageing clocks and nonlinear multiomic ageing profiles offer complementary tools for identifying patients whose biological organ age is discordant with chronological age, although these clocks still need to be linked to senescence causality and therapeutic reversibility (Shen et al., 2024; Wang et al., 2026).

Because these measurements vary across tissues and platforms, study-specific cut-offs should not be mistaken for clinical diagnoses. Findings should instead be reported in evidence tiers. A senescence-associated state requires orthogonal same-cell markers. A persistent senescent niche additionally requires quantitative spatial enrichment and an identifiable maintenance mechanism. The designation senescent repair memory should be reserved for studies that also demonstrate persistence beyond the expected repair phase and functional coupling or reversibility. These are proposed reporting categories rather than validated diagnostic classes. Longitudinal studies should report continuous trajectories and effect sizes, while exploratory cut-offs should be labelled as cohort- and assay-specific and externally validated before clinical application.

6. Therapeutic and clinical translation

These evidence principles lead directly to a therapeutic implication: treatment should target a tissue's failure to exit an injury-induced programme rather than senescence in the abstract (McHugh et al., 2025; Ogrodnik et al., 2024). Senolytics remove senescent cells, whereas senomorphics suppress deleterious SASP without necessarily eliminating the cells (McHugh et al., 2025; Zheng et al., 2024). Neither approach is sufficient if tissue resolution is not restored (Martin et al., 2024; Mylonas and Ferenbach, 2024). Human evidence remains preliminary: a small randomized phase I IPF pilot of intermittent dasatinib plus quercetin was feasible and generally tolerated but was underpowered for efficacy and showed no meaningful between-group differences in pulmonary or physical function (Nambiar et al., 2023). A five-participant phase I Alzheimer disease study supported feasibility and CNS penetration but did not establish cognitive or neuroimaging benefit (Garbarino et al., 2025). These studies provide early safety, feasibility and pharmacodynamic information rather than evidence of disease modification. Major unresolved bottlenecks include heterogeneous senolytic targets, uncertain tissue exposure, absence of validated target-engagement assays, limited knowledge of the optimal treatment window, variable dependence of different senotypes on pro-survival pathways and the possibility that short-term depletion is followed by repopulation of the same pathological niche. Trials should therefore incorporate tissue- or organ-specific pharmacodynamics, functional endpoints, delayed safety assessment and prespecified stopping rules rather than infer benefit from circulating marker reduction alone. SGLT2 inhibitors and GLP-1-based therapies should be described primarily as metabolic, haemodynamic and parenchymal disease-modifying drugs that may secondarily influence senescence-related states; their classification as senotherapeutics remains unproven (Katsuumi et al., 2024;

Yesilyurt-Dirican et al., 2025; Drucker, 2025).

Within this broader strategy, immune-guided senotherapy may be useful when senescent cells persist through checkpoint-mediated immune escape (Majewska et al., 2024; Iltis et al., 2025). Strategies that enhance natural killer-cell, macrophage or T-cell clearance, block senescence-associated immune checkpoints or engineer senolytic immune cells remain predominantly preclinical (Ming et al., 2025; Amor et al., 2024). CAR T-cell approaches require stringent control of antigen heterogeneity, on-target off-tissue recognition, cellular persistence, reversibility and cytokine-mediated or other immune toxicity. A target such as uPAR may identify relevant populations in one disease stage yet also be expressed by activated non-senescent cells in another, making antigen validation and normal-tissue mapping essential. Spatially targeted delivery may reduce systemic exposure, but nanoparticles, prodrugs, extracellular vesicles and niche-specific ligands introduce additional uncertainties related to biodistribution, incomplete niche coverage, manufacturing reproducibility and long-term retention (Magkouta et al., 2025; Gu et al., 2024; Li et al., 2025). Table 2 summarizes therapeutic strategies, candidate biomarkers, principal risks and design implications. Clinical translation will require organ-, stage- and senotype-specific pharmacodynamic evidence, demonstration that adaptive repair is preserved and functional benefit beyond marker reduction.

Regardless of modality, a key unresolved question is whether senescent repair memory remains reversible after spatial niches and epigenetic fixation are established (Muto et al., 2024; Vu et al., 2026). Reversibility is unlikely to be binary: suppression of SASP may reduce inflammation without removing the cellular reservoir, senolysis may remove the reservoir while leaving a mechanically fixed matrix, and metabolic correction may fail if immune tolerance has already stabilized the niche. Studies should therefore distinguish molecular, cellular, architectural and functional reversal and determine whether each persists after treatment withdrawal. If reversibility differs among tissues and disease stages, senotherapy may need to be combined sequentially with regenerative, antifibrotic, immunomodulatory or metabolic interventions rather than administered as a single universal strategy (McHugh et al., 2025; Abdellatif et al., 2025).

This uncertainty should shape trial design. Clinical translation can be structured around the evidence tiers in 5 and Table 1 as a sequence for eligibility assessment and endpoint selection rather than as a direct clinical diagnostic scheme. Trials should identify the initiating injury and candidate cellular compartment, document the strongest evidence tier supported at baseline, measure early target engagement within that compartment and require later improvement in tissue architecture or organ function. A reduction in senescence markers without architectural or functional benefit should be reported as pharmacodynamic activity rather than successful resolution (Ogrodnik et al., 2024; Zheng et al., 2024).

Patient selection should likewise be based on unresolved repair biology rather than chronological age or senescence markers alone (Shen et al., 2024; Wen, 2025). A repair-memory-informed trial could enrich for evidence of persistent senescence, active niche pathology and residual reversibility (McCallion et al., 2024; Wen, 2025). In CKD, enrichment might combine eGFR slope, albuminuria, tubular injury markers, inflammatory proteomics, senescence-associated proteins and imaging or biopsy-based fibrosis metrics (Zhang et al., 2024a; McCallion et al., 2024). Comparable disease-specific combinations should be defined for cardiac remodelling, pulmonary function, liver fibrosis, neurodegenerative progression and cartilage loss. Trial design should include a baseline measure of the proposed senescent compartment, an early target-engagement endpoint, a later organ-level efficacy endpoint and follow-up after treatment withdrawal to determine whether the niche re-emerges. Circulating SASP proteins or tissue senescence scores can provide pharmacodynamic information, but clinically meaningful resolution requires stabilization of organ function, improved regenerative capacity, reduced fibrosis or diminished inflammatory recurrence

Table 2
Therapeutic strategies in a repair-resolution framework.

Strategy	Best-fit repair-memory state	Biomarkers or endpoints	Main risk	Design implication
Senolytics	Established pathological senescent niches with low adaptive repair value.	Reduced senescent-cell burden, confirmed target engagement and improvement in organ architecture or function.	Loss of beneficial repair cells, systemic toxicity and incomplete target specificity.	Require compartment-specific target engagement, intermittent or stage-defined dosing, local delivery when feasible and delayed safety follow-up.
Senomorphics	Persistent SASP-driven inflammation where cell removal is unsafe.	Lower SASP proteins, reduced paracrine senescence, improved niche behaviour and preserved repair function.	The pathogenic cellular reservoir may remain and re-establish the niche after treatment withdrawal.	Separate pharmacodynamic SASP suppression from organ benefit and combine with clearance or regenerative strategies only when justified.
Metabolic senomorphics	Metabolically fixed senotypes driven by hypoxia, lipotoxicity, uremia or mitochondrial stress.	Mitochondrial, redox, lipid-peroxidation, NAD ⁺ or AMPK/mTOR target-engagement readouts together with organ-level outcomes.	Organ-specific response variability and uncertain attribution when the intervention has multiple established actions.	Select patients by senotype and tissue context. Do not classify established metabolic therapies as senotherapeutics without cell-specific validation.
Immune-guided senotherapy	Senescent-cell retention through checkpoint signalling or impaired efferocytosis.	NK-cell, macrophage, T-cell and checkpoint signatures, restored clearance and organ-level benefit.	On-target off-tissue recognition, cytokine-mediated toxicity, prolonged immune activation and impaired wound repair.	Map antigen expression in normal tissues, use controllable or reversible platforms and include long-term immune and safety monitoring.
Spatially targeted delivery	Fibrotic, vascular, pulmonary, renal or cartilage niches where systemic exposure is risky.	Niche-local drug exposure, spatial marker reduction, confirmed biodistribution and tissue-specific functional benefit.	Incomplete coverage of diffuse disease, uncertain biodistribution and manufacturing variability.	Validate tissue exposure and spatial coverage before efficacy testing. Use nanoparticles, prodrugs, engineered cells or tissue-selective ligands only with organ-specific safety assessment.
Combination resolution therapy	Late disease sustained by metabolic fixation, immune retention and matrix persistence together.	Composite endpoints combining target engagement, spatial niche change, organ function and persistence after treatment withdrawal.	Complex safety, attribution, sequence dependence and cumulative toxicity.	Prespecify treatment sequence and stopping rules. Evaluate each component and the combination against repair, immune, architectural and functional endpoints.

(Mylonas and Ferenbach, 2024; Abdellatif et al., 2025).
Once patients are selected, safety monitoring must account for the roles of senescent cells in tumour suppression, wound repair and tissue remodelling (Huang et al., 2022; McHugh et al., 2025). Indiscriminate

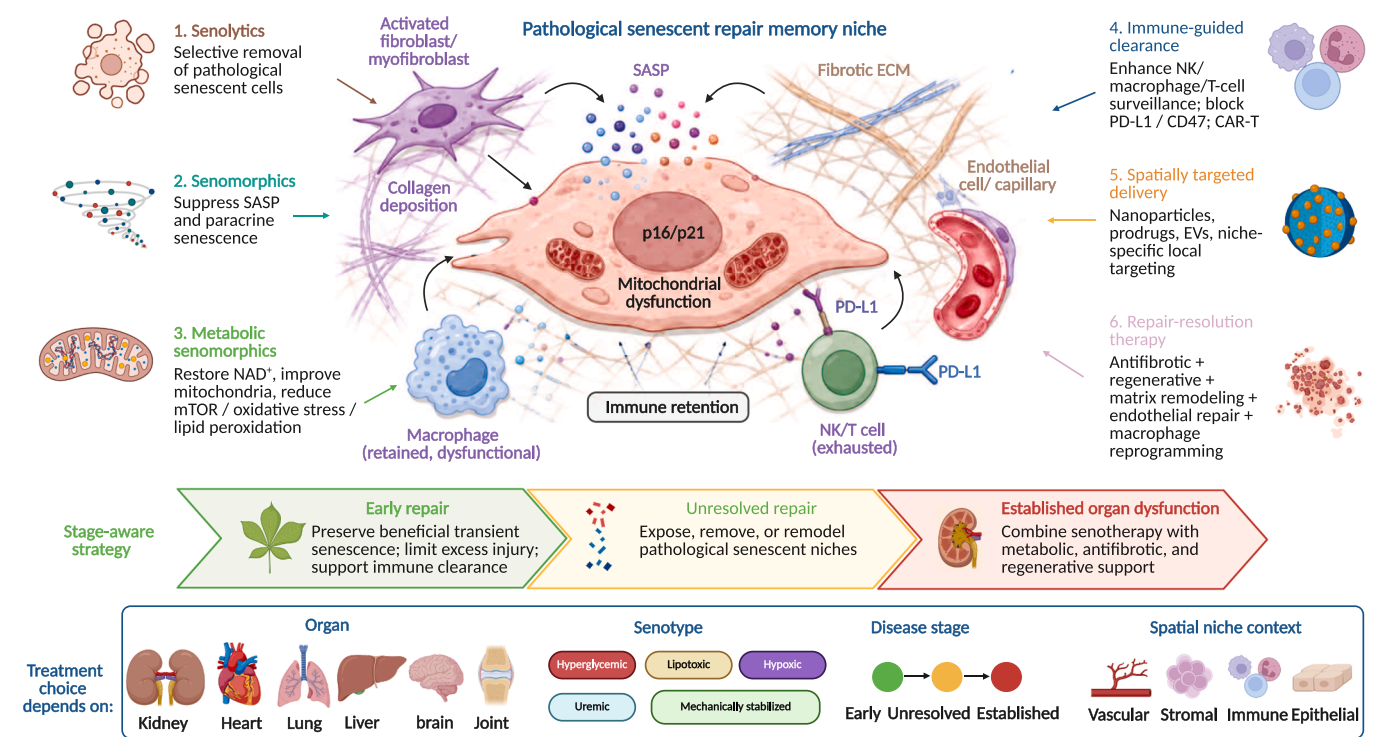


Fig. 3. Therapeutic strategies for targeting pathological senescent repair memory. Senescent repair memory requires stage- and niche-aware intervention rather than indiscriminate suppression of senescence. A pathological repair-memory niche may contain p16- or p21-positive senescent cells, mitochondrial dysfunction, persistent SASP signalling, activated fibroblasts or myofibroblasts, fibrotic ECM, endothelial dysfunction, retained macrophages and exhausted NK or T cells. Therapeutic strategies include senolytics to remove pathological senescent cells, senomorphics to suppress SASP and paracrine senescence, metabolic senomorphics to restore NAD⁺, improve mitochondrial function and reduce mTOR activation, oxidative stress or lipid peroxidation, immune-guided approaches to enhance macrophage, NK-cell or T-cell clearance, spatially targeted delivery using nanoparticles, prodrugs, extracellular vesicles or tissue-selective ligands, and combination strategies incorporating antifibrotic, regenerative, matrix-remodelling, endothelial-repair or macrophage-reprogramming interventions. Treatment choice should be guided by organ context, senotype, disease stage and spatial niche composition. Early repair may require preservation of beneficial transient senescence, whereas unresolved repair may require selective clearance or niche remodelling; established organ dysfunction may additionally require regenerative or organ-protective support. SASP, senescence-associated secretory phenotype; ECM, extracellular matrix; NK, natural killer; EVs, extracellular vesicles.

or poorly timed elimination could remove beneficial repair cells, delay epithelial closure, impair stromal support, worsen tissue fragility or produce systemic off-target toxicity. Immune-guided strategies add risks of prolonged immune activation, cytokine-mediated injury and loss of antigen-positive non-senescent cells, whereas systemic senolytics may have tissue-specific toxicities determined by their molecular targets. These risks favour stage-defined patient selection, intermittent dosing, local delivery, reversible platforms or prodrug activation where justified (Magkouta et al., 2025; Li et al., 2025). Early-phase trials should include wound-healing surveillance, organ-specific toxicity, immune monitoring and long-term assessment of tumour and infection risk. Combination therapy may be necessary when several maintenance axes coexist, but it increases attribution and safety complexity; rational combinations should therefore be evaluated against organ function, wound repair, immune toxicity and tissue architecture, not only senescence-marker reduction.

The balance between efficacy and safety will vary by disease stage. During early repair, treatment may need to preserve beneficial senescence while limiting excessive metabolic injury and supporting immune clearance of irreversibly damaged cells (Wang et al., 2024a; Majewska and Krizhanovsky, 2025). During unresolved repair, intervention may instead target mitochondrial dysfunction, lipid peroxidation, immune-checkpoint signalling, matrix stiffening or epigenetic fixation (Majewska et al., 2024; Katsuumi et al., 2024). In established organ failure, complete reversal may be unrealistic, and senotherapy may need to be paired with antifibrotic, regenerative, metabolic or organ-protective strategies to slow progression (Martin et al., 2024; McHugh et al., 2025). These proposals remain translational hypotheses rather than established treatment algorithms (Fig. 3).

This stage-specific logic differs from conventional anti-ageing strategies because it treats senescence as a temporally changing repair programme. The relevant clinical question is not simply whether a drug lowers a senescence marker, but whether it engages the intended target without disrupting adaptive repair and improves a prespecified organ-level outcome (Suryadevara et al., 2024; Ogrodnik et al., 2024).

The same logic must also be adapted to each organ because the cellular compartments, maintenance mechanisms and measurable outcomes differ across diseases. CKD provides a tractable example: progression can be quantified through functional decline, histological fibrosis, tubular injury markers and repeated metabolic stress (Zhang et al., 2024a; Hejazian et al., 2024). Injured proximal tubules may initiate unresolved repair, whereas podocytes, endothelial cells, macrophages, fibroblasts and pericytes can amplify or stabilize the niche (Muto et al., 2024; Chen et al., 2024b; Ma et al., 2025). Candidate interventions must therefore demonstrate renal-compartment specificity and preservation of protective repair (Zhang et al., 2024b; Russo et al., 2024).

In the kidney, for instance, immune retention should be assessed beyond generic inflammation because macrophage phenotypes, T-cell exhaustion, impaired efferocytosis and immune-checkpoint pathways may determine whether senescent tubules are removed or retained (Majewska and Krizhanovsky, 2025; Wu et al., 2024a). Informative designs would combine tubular senescence reporters with immune-cell perturbation and spatial analysis of tubule-macrophage-fibroblast communication (Shao et al., 2024a; Wu et al., 2024a). These experiments should distinguish a dynamic repair state from a static ageing biomarker and link cell-state changes to renal function and tissue architecture (Suryadevara et al., 2024; Zhang et al., 2024a).

The same compartment-resolved approach should be applied in other organs to avoid treating the kidney as the default model for all chronic disease. In the heart, candidate reservoirs include perivascular cells, endothelial cells and fibroblasts, with ventricular remodelling and vascular stiffness as functional outcomes. In the lung, alveolar epithelial-fibroblast niches should be linked to compliance, gas exchange and fibrosis progression; in the liver, stellate-cell and hepatocyte states should be related to spatial fibrosis and metabolic function; in the brain,

glial states require regional, synaptic and behavioural validation; and in cartilage, chondrocyte and matrix states should be linked to structural loss and pain-related function. Human tissue atlases, circulating biomarkers, spatial omics and selective perturbation will be required to determine which compartment is causal in each disease and whether targeting it preserves protective repair elsewhere (Suryadevara et al., 2024; Ogrodnik et al., 2024).

This principle implies distinct trial designs for cardiovascular, pulmonary and hepatic disease. In HFpEF, pericyte, endothelial, fibroblast and macrophage states should be linked to diastolic function, exercise capacity and fibrosis, while vascular studies should use endothelial function, arterial stiffness or plaque imaging rather than circulating SASP alone (Silva et al., 2025; Vanicek et al., 2025; Potenza et al., 2025). Pulmonary trials should distinguish AT2-fibroblast niches in pulmonary fibrosis from airway epithelial propagation in COPD and pair local target engagement with forced vital capacity, gas transfer, imaging or exacerbation outcomes (Parimon et al., 2023; Devulder et al., 2025; Guo-Parke et al., 2025; Li et al., 2024b; Wang et al., 2024b; Nambiar et al., 2023). In MASLD, stellate-cell and hepatocyte states should be mapped against spatial fibrosis, metabolic function and fibrosis regression, with follow-up after weight loss or metabolic control to determine whether the niche persists (Vu et al., 2026; Yashaswini et al., 2024; Kiourtis et al., 2024; Yang et al., 2025).

By contrast, neurological and joint studies face different accessibility constraints. Brain trials may require cerebrospinal-fluid or imaging pharmacodynamic markers, but these measures should be anchored to regional glial states and to cognitive, synaptic or motor outcomes rather than interpreted as direct evidence of senescent-cell depletion (Millet et al., 2024; Avey et al., 2025; Wang et al., 2024c; de Luzy et al., 2024; Rim et al., 2024; Carling et al., 2024; Alshaebi et al., 2025; Caceres-Palomo et al., 2025; Rohden et al., 2025; Gruel et al., 2024; Richardson and Richardson, 2024; Garbarino et al., 2025). Osteoarthritis studies can obtain synovial fluid and operative tissue more readily, allowing local target engagement to be linked to quantitative cartilage imaging, joint structure, pain and function (Huang et al., 2024; Singla et al., 2026; Shao et al., 2024b; Lin et al., 2024a, 2024b; Wang et al., 2024d; Jiang et al., 2024; Zhao et al., 2025b; Maurer et al., 2025; Feng et al., 2024; Li et al., 2025; Zhang et al., 2025). These organ-specific requirements reduce the risk that a renal biomarker workflow is transferred uncritically to diseases with different repair kinetics and tissue access.

7. Future perspectives

These translational challenges define the next research priorities. The field should now move from association to direct testing of the model. The central question is no longer simply whether senescent cells are present in chronic disease, but when, where and why the senescence programme fails to resolve (Martin et al., 2024; Suryadevara et al., 2024; Ogrodnik et al., 2024). This requires longitudinal sampling, spatial mapping, single-cell multiomics and perturbation of immune clearance, metabolic fixation and niche architecture (Muto et al., 2024; SenNet Consortium, 2022). A first priority is to define temporal trajectories of adaptive and maladaptive senescence during injury, repair and chronic progression (Yeh et al., 2024; Muto et al., 2024). Prospective studies should prespecify the expected repair window, sample before and after that boundary, identify the same cell lineage across time and relate persistence to subsequent organ decline. Stage-specific perturbation should then determine whether the implicated population is beneficial during early repair but pathogenic after resolution has failed. Such experiments would establish the therapeutic interval in which senescence shifts from supporting repair to sustaining dysfunction and would clarify when senolysis, senomorphism or niche remodelling is most likely to be beneficial or harmful (Mylonas and Ferenbach, 2024; McHugh et al., 2025).

Beyond temporal mapping, a second priority is to classify metabolic

senotypes across diseases and organs (Vu et al., 2026; Monroe et al., 2025). Hyperglycaemic, lipotoxic, hypoxic, uremic, proteotoxic and mechanically stabilized senotypes may require different interventions even when they share p16, p21 or SASP markers (Xie et al., 2024; Fu et al., 2024). A third priority is to define immune escape pathways in non-malignant senescent cells and determine whether checkpoint blockade, engineered immune cells or innate immune stimulation can safely restore clearance (Majewska et al., 2024; Ming et al., 2025). The challenge will be to enhance senescent-cell removal without provoking autoimmunity, impairing wound repair or destabilizing tissues that rely on transient senescence for regeneration (Huang et al., 2022; Majewska and Krizhanovsky, 2025).

A further priority is to test whether senescent niches can be remodelled without complete cell elimination (Tsukui et al., 2024; Vu et al., 2026). Matrix-softening therapies, macrophage reprogramming, metabolic rewiring, endothelial repair and progenitor niche restoration could convert pathological repair memory into a more permissive regenerative environment (Gu et al., 2024; Abdellatif et al., 2025).

8. Conclusion

Overall, the evidence reviewed here suggests that senescence should not be treated simply as a marker of ageing or a pathogenic cell state that should be removed from chronically diseased tissues. The senescent repair memory model offers a more specific interpretation: senescence becomes most damaging when a normally transient repair programme fails to terminate and is stabilized by metabolic stress, immune retention, chromatin states and spatial niche architecture. The concept is an organizing and testable framework, not a claim that all chronic disease is caused by senescence or that all senescent cells should be eliminated. Its value will depend on cell-resolved biomarkers, longitudinal human studies, causal perturbations and clinically meaningful resolution endpoints. Across organ systems, this approach shifts attention from measuring senescence burden to defining and therapeutically modifying failed repair trajectories.

Abbreviations

4-HNE, 4-hydroxynonenal; AhR, aryl hydrocarbon receptor; AKI, acute kidney injury; AMPK, AMP-activated protein kinase; APOE4/4, apolipoprotein E ϵ 4/ ϵ 4 genotype; AT2, alveolar type 2; ATP, adenosine triphosphate; BCL-2, B-cell lymphoma 2; BCL-xL, B-cell lymphoma-extra large; CAR T cells, chimeric antigen receptor T cells; CD8⁺ T cells, cluster of differentiation 8-positive T cells; CD47, cluster of differentiation 47; cGAS-STING, cyclic GMP-AMP synthase–stimulator of interferon genes; CKAP4, cytoskeleton-associated protein 4; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; CoA, coenzyme A; CXCL1, C-X-C motif chemokine ligand 1; CXCL8, C-X-C motif chemokine ligand 8; DAMPs, damage-associated molecular patterns; DNA, deoxyribonucleic acid; ECM, extracellular matrix; eGFR, estimated glomerular filtration rate; ER, endoplasmic reticulum; EVs, extracellular vesicles; Fe²⁺, ferrous iron; GD3, disialoganglioside GD3; GLP-1, glucagon-like peptide-1; H2AX, histone H2AX; γ -H2AX, phosphorylated histone H2AX; HDAC3, histone deacetylase 3; HFPEF, heart failure with preserved ejection fraction; HGF, hepatocyte growth factor; HLA-E, human leukocyte antigen E; HNRNP, heterogeneous nuclear ribonucleoprotein D; IFN, interferon; IGFBP2, insulin-like growth factor-binding protein 2; IL-1 α / β , interleukin-1 alpha/beta; IL-6, interleukin-6; IL-23R, interleukin-23 receptor; iPSC, induced pluripotent stem cell; Lonp1, Lon peptidase 1; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; miR, microRNA; MMPs, matrix metalloproteinases; mRNA, messenger RNA; mtDNA, mitochondrial DNA; mtRNA, mitochondrial RNA; mTOR, mechanistic target of rapamycin; NAD⁺, oxidized nicotinamide adenine dinucleotide; NK cells, natural killer cells; NKG2A, natural killer group 2 A; O₂, oxygen; OXPHOS,

oxidative phosphorylation; p16, cyclin-dependent kinase inhibitor 2 A; p21, cyclin-dependent kinase inhibitor 1 A; PDGF, platelet-derived growth factor; PD-L1, programmed death ligand 1; PDZK1, PDZ domain-containing 1; PROTAC, proteolysis-targeting chimera; PTX3, pentraxin 3; RNA, ribonucleic acid; ROS, reactive oxygen species; S100A8, S100 calcium-binding protein A8; SA- β -galactosidase, senescence-associated β -galactosidase; SASP, senescence-associated secretory phenotype; SGLT2, sodium-glucose cotransporter 2; SIRP α , signal regulatory protein alpha; Sirt4, sirtuin 4; SIRT6, sirtuin 6; TGF- β , transforming growth factor beta; uPAR, urokinase plasminogen activator receptor.

Author contributions

H.Hu and R.Ji collected and evaluated the literature, drafted the manuscript and created the figures. J. Hu conceived the review framework, supervised the work and critically revised the manuscript. All authors read and approved the final manuscript.

Informed consent statement

Not applicable.

Institutional review board statement

Not applicable.

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

The authors declare that no generative AI or AI-assisted technologies were used in the preparation of this manuscript.

Funding

This work was supported by grants from the National Natural Science Foundation of China (82300767 to J.Hu).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Figs. 1–3 were created with BioRender.com.

Data Availability

Data sharing does not apply to this article as no datasets were generated during the current study.

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